Practice Worksheet 2: Conditionals

CORE 103 Computational Thinking

Fall 2026

Instructions

- This PW is **UNGRADED**. However, it is highly encouraged that you attempt and submit it to get feedback from your section tutor.
- This PW must be submitted online via CANVAS. You are required to submit a zip file that contains all the `.py` files.
- The zip file should be named as `PW2_aa123456.zip` where `aa123456` will be replaced with your student id.

Exercises

1. Board Game Critical Roll

This question should be implemented in the file **q1.py**.

Two friends are waiting for their group project meeting to start, so they play a quick board-game round. Each friend rolls one die numbered from 1 to 20.

In their house rules, a round becomes exciting enough to call "CRIT" only when the two rolls together cross the critical limit of 20. Given the two rolls `x` and `y`, decide whether the round is a critical roll.

Constraints

- $x, y \in \mathbb{N}$
- $1 \leq x, y \leq 20$

Interaction

The input contains two space-separated integers: `x` and `y`.

The output must be "CRIT" if the sum is greater than 20, and "NO CRIT" otherwise.

Sample

Input	Output
10 2	NO CRIT
12 13	CRIT

In the first case, $10 + 2 = 12$, which is not greater than 20, so the output is "NO CRIT".

In the second case, $12 + 13 = 25$, which is greater than 20, so the output is "CRIT".

Exercise

Indicate the outputs for the given inputs in **q1.py**.

Input	Output
6 17	
10 10	
7 12	

Python Code

Implement the Python code in **q1.py**.

2. Match Score Check

This question should be implemented in the file **q2.py**.

A student is following a live campus quiz match between two teams. At one moment, the score update shows $a : c$. A few minutes later, a friend sends another update showing $b : d$.

In a real match, a team's score can stay the same or increase, but it cannot go backwards. Decide whether the second update could be a valid later score.

Constraints

- $a, c, b, d \in \mathbb{Z}_{\geq 0}$
- $0 \leq a, c, b, d \leq 20$

Interaction

The input contains four space-separated integers: `a`, `c`, `b`, and `d`.

The output must be "Possible" if the later score can happen, and "Impossible" otherwise.

Sample

Input	Output
1 5 3 5	Possible
3 4 2 6	Impossible

In the first case, the score can move from 1 : 5 to 3 : 5 because Team 1 can score two more points and Team 2's score does not decrease.

In the second case, the score cannot move from 3 : 4 to 2 : 6 because Team 1's score would decrease.

Exercise

Indicate the outputs for the given inputs in **q2.py**.

Input	Output
2 2 2 2	
0 1 8 0	
9 9 7 13	

Python Code

Implement the Python code in **q2.py**.

3. Exam Preparation Plan

This question should be implemented in the file **q3.py**.

An 18-year-old first-year student is trying to plan exam preparation without ignoring classes, commute, and family responsibilities. The exam is in `n` days, and the syllabus has `x` pages.

The student can realistically study at most `y` pages per day. Decide whether the student can finish the syllabus before the exam.

Constraints

- $x, n, y \in \mathbb{N}$
- $1 \leq x \leq 500$
- $1 \leq n \leq 60$
- $1 \leq y \leq 50$

Interaction

The input contains three space-separated integers: `x`, `n`, and `y`.

The output must be "YES" if the student can finish the syllabus, and "NO" otherwise.

Sample

Input	Output
30 5 5	NO
90 10 10	YES

In the first case, the student can cover at most $5 \times 5 = 25$ pages, which is not enough for 30 pages.

In the second case, the student can cover at most $10 \times 10 = 100$ pages, which is enough for 90 pages.

Exercise

Indicate the outputs for the given inputs in **q3.py**.

Input	Output
99 11 9	
45 6 7	
5 1 5	

Python Code

Implement the Python code in **q3.py**.

4. Commute Cost Choice

This question should be implemented in the file **q4.py**.

A student is comparing two commute options for the daily route to university. Option 1 travels `d1` km using one litre of fuel, and its fuel costs `p1` rupees per litre. Option 2 travels `d2` km using one litre of fuel, and its fuel costs `p2` rupees per litre.

For the same route, the better option is the one that costs less to travel each kilometre. Determine which option is cheaper.

Constraints

- $d1, d2, p1, p2 \in \mathbb{N}$
- $1 \leq d1, d2 \leq 100$
- $1 \leq p1, p2 \leq 1000$

Interaction

The input contains four space-separated integers: `d1`, `d2`, `p1`, and `p2`.

The output must be -1 if option 1 is cheaper, 1 if option 2 is cheaper, and 0 if both cost the same.

Sample

Input	Output
10 5 3 20	-1
1 5 3 2	1

In the first case, option 1 costs $3/10 = 0.3$ rupees per km, while option 2 costs $20/5 = 4$ rupees per km. Option 1 is cheaper, so the output is -1.

In the second case, option 1 costs $3/1 = 3$ rupees per km, while option 2 costs $2/5 = 0.4$ rupees per km. Option 2 is cheaper, so the output is 1.

Exercise

Indicate the outputs for the given inputs in **q4.py**.

Input	Output
7 2 7 2	
10 3 5 12	
12 3 2 9	

Python Code

Implement the Python code in **q4.py**.

5. University Trip Packing Check

This question should be implemented in the file **q5.py**.

Amna and her friends are leaving for a university society trip. The bus team allows each student to place exactly two bags in the luggage compartment and keep one bag with them on the seat. Amna has three bags weighing `A`, `B`, and `C` kg.

The two bags placed in the luggage compartment must have a combined weight of at most `D` kg. The bag kept with Amna must weigh at most `E` kg. She can choose which two bags go in the luggage compartment and which one stays with her.

Given `A`, `B`, `C`, `D`, and `E`, determine whether Amna can arrange her bags without paying an extra luggage fee.

Constraints

- $A, B, C, D, E \in \mathbb{N}$
- $1 \leq A, B, C \leq 10$
- $15 \leq D \leq 20$
- $5 \leq E \leq 10$

Interaction

The input comprises a single line containing 5 space-separated integers denoting the values of `A`, `B`, `C`, `D`, and `E` respectively.

The output should contain a single word, YES, if Amna can arrange the bags without paying an extra fee, and NO if she cannot.

Sample

Input	Output
1 1 1 15 5	YES
8 7 6 15 5	NO

In the first case, $(A, B, C, D, E) = (1, 1, 1, 15, 5)$. Amna can place the first and second bags in the luggage compartment because their total weight is at most 15 kg, and she can keep the third bag with her because it is at most 5 kg. The output is YES.

In the second case, $(A, B, C, D, E) = (8, 7, 6, 15, 5)$. Every possible choice either makes the luggage-compartment pair too heavy or leaves a carry-on bag above the allowed limit. The output is NO.

Exercise

Indicate the outputs for the given inputs in **q5.py**.

Input	Output
8 5 7 15 6	
8 8 9 16 5	
10 10 10 20 10	

Python Code

Implement the Python code in **q5.py**.

6. Library Book Pickup Window

This question should be implemented in the file **q6.py**.

The university library is holding reserved books for students. A student reserved a book `d` days ago. The pickup window is from day `a` through day `b`, inclusive.

Decide whether the student has come too early, too late, or during the correct pickup window.

Constraints

- $d, a, b \in \mathbb{N}$
- $1 \leq d \leq 365$
- $1 \leq a \leq b \leq 365$

Interaction

The input contains three space-separated integers: `d`, `a`, and `b`.

The output must be "Too Early", "Too Late", or "Pickup Allowed".

Sample

Input	Output
8 8 12	Pickup Allowed
14 2 10	Too Late

In the first case, `d`=8 and the valid pickup window is from day 8 to day 12, so the student can pick up the book now.

In the second case, `d`=14 and the valid pickup window is from day 2 to day 10, so it is too late.

Exercise

Indicate the outputs for the given inputs in **q6.py**.

Input	Output
4 5 6	
8 10 12	
9 2 21	

Python Code

Implement the Python code in **q6.py**.

7. Part-Time Pay Calculator

This question should be implemented in the file **q7.py**.

A university student takes paid event shifts for a campus event company. The company calculates the event stipend using a base amount plus two support amounts. The support rule is different for smaller and larger event contracts.

For smaller event contracts below Rs. 1500, the travel support is 10% of the base amount and the event bonus is 90% of the base amount. Otherwise, the travel support is fixed at Rs. 500 and the event bonus is 98% of the base amount. Given `s`, calculate the total stipend.

Constraints

- $s \in \mathbb{N}$
- $500 \leq s \leq 50000$

Interaction

The input contains a single integer: `s`, the base amount.

The output must contain a float denoting the total stipend.

Sample

Input	Output
1203	2406.0
10042	20383.16

In the first case, `s`=1203, which is less than 1500. The travel support is 120.3 and the event bonus is 1082.7. The total stipend is $1203 + 120.3 + 1082.7 = 2406.0$.

In the second case, `s`=10042, which is at least 1500. The travel support is 500 and the event bonus is 9841.16. The total stipend is $10042 + 500 + 9841.16 = 20383.16$.

Exercise

Indicate the outputs for the given inputs in **q7.py**.

Input	Output
1312	
2300	
1899	

Python Code

Implement the Python code in **q7.py**.

8. Campus Receipt Code

This question should be implemented in the file **q8.py**.

A group of students is ordering snacks from the campus cafe for a small birthday surprise. The cafe billing system prints a special receipt code only when the final bill amount falls in the five-digit range.

The group buys `n` snack boxes, and each box costs `p` rupees. Decide whether the final bill amount can be printed as a valid receipt code. A valid amount is between 10000 and 99999 inclusive.

Constraints

- $n, p \in \mathbb{N}$
- $1 \leq n \leq 500$
- $1 \leq p \leq 1000$

Interaction

The input contains two space-separated integers: `n` and `p`.

The output must be "YES" if the total bill is a valid 5-digit receipt code, and "NO" otherwise.

Sample

Input	Output
25 785	YES
402 11	NO

In the first case, `n`=25 and `p`=785. The total bill is $25 \times 785 = 19625$, which is a valid 5-digit code, so the output is "YES".

In the second case, `n`=402 and `p`=11. The total bill is $402 \times 11 = 4422$, which is not a 5-digit code, so the output is "NO".

Exercise

Indicate the outputs for the given inputs in **q8.py**.

Input	Output
100 100	
33 12	
130 120	

Python Code

Implement the Python code in **q8.py**.

9. Campus Step Challenge

This question should be implemented in the file **q9.py**.

During orientation week, Hilda joins a funny campus step challenge with her friends. The challenge app gives her a repeated movement pattern: on one move she walks 3 steps forward, and on the next move she moves 1 step backward because the group stops and pulls back for a photo. This forward-backward pattern keeps repeating.

After making `k` total movements, Hilda wants to know how far ahead she is from the starting point. Help Hilda find her final position relative to where she began.

Constraints

- $k \in \mathbb{N}$
- $1 \leq k \leq 10^5$

Interaction

The input comprises a single line containing an integer denoting the value of `k`.

The output should contain a single integer denoting the number of steps Hilda is ahead of her starting point at the end of `k` movements.

Sample

Sample

Input	Output
5	7
4	4

In the first case, `k`=5. The number of steps can be calculated as $(3 - 1 + 3 - 1 + 3) = 7$.

In the second case, `k`=4. The number of steps can be calculated as $(3 - 1 + 3 - 1) = 4$.

Exercise

Indicate the outputs for the given inputs in **q9.py**.

Input	Output
12	
8	
31	

Python Code

Implement the Python code in **q9.py**.

10. Player Comparison

This question should be implemented in the file **q10.py**.

Two students are comparing their campus cricket performance after a friendly match. For each player, three statistics are recorded: runs, wickets, and catches.

A player is considered better overall if they are ahead in more of the three statistics. The first input line contains Player A's statistics, and the second input line contains Player B's statistics. Determine which player is better.

Constraints

- $ra, rb \in \mathbb{Z}_{\geq 0}$
- $wa, wb, ca, cb \in \mathbb{Z}_{\geq 0}$
- $0 \leq ra, rb \leq 500$
- $0 \leq wa, wb, ca, cb \leq 20$

Interaction

The first line contains `ra`, `wa`, and `ca`. The second line contains `rb`, `wb`, and `cb`.

The output must be A if Player A is better, B if Player B is better, and TIE if neither is better.

Sample

Input	Output
0 1 2 2 3 4	B
10 0 10 0 10 0	A

In the first case, Player B is better in all three statistics.

In the second case, Player A is better in two out of the three statistics.

Exercise

Indicate the outputs for the given inputs in **q10.py**.

Input	Output
34 5 6 23 5 6	
123 14 5 323 1 6	
199 14 16 199 13 17	

Python Code

Implement the Python code in **q10.py**.

Rough Work